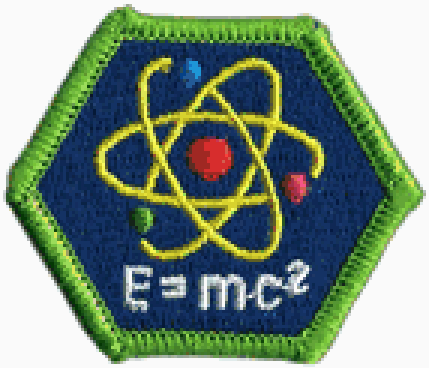
SCOUTS BSA TEST LAB SERIES



PHYSICS



Scouting America

Scouts BSA Test Lab Resource Guide: Physics
Requirement 1 ? Safety

Safety. Before conducting any experiments, explain to your counselor the safety precautions and personal protective equipment necessary for working with physics-related equipment.

Official Study & Counselor Discussion Points:

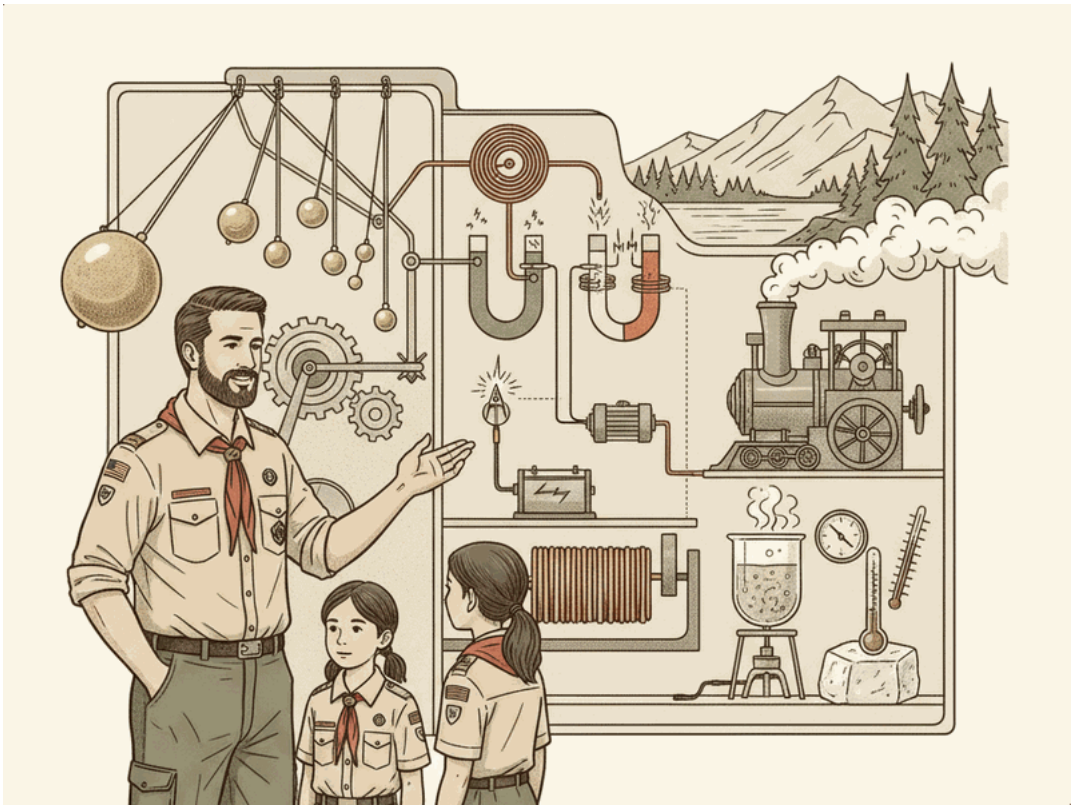


Scouts BSA Test Lab Resource Guide: Physics
Requirement 2 ? What is Physics?

What is Physics? Define physics in your own words and discuss with your counselor various areas of physics.

Official Study & Counselor Discussion Points:

? Resource: What is Physics? (video) (https://youtu.be/yWMKYID5fr8)
? Resource: Why study physics? (website) (https://www.aps.org/careers/advice/why-study-physics)
? Resource: The Map of Physics (video) (https://youtu.be/ZihywtixUYo)



Scouts BSA Test Lab Resource Guide: Physics
Requirement 3 ? Fundamentals of Physics

Fundamentals of Physics. Explain these key physics concepts: force, motion, energy, and waves. Provide real-world examples demonstrating each principle.

Official Study & Counselor Discussion Points:
? Resource: All physics explained in 15 minutes (worth remembering) (video) (https://youtu.be/TTHazQeM8v8)







Scouts BSA Test Lab Resource Guide: Physics
Requirement 6 ? The Future of Physics

The Future of Physics. Physicists try to explain the universe and apply these concepts to novel technologies. Discuss one option with your counselor: (A) Option A: Unsolved Mysteries of Physics. Discuss the following with your counselor: (a) Dark Matter & Dark Energy. Scientists know it exists?but what is it? (b) Black Holes & Wormholes. Can we harness their energy or travel through them? (B) Option B: Quantum Mechanics & Computing. Discuss the following with your counselor: (a) Quantum Computers. How can super fast computers be made of atoms instead of microchips? (b) Quantum Entanglement. Can we communicate instantly across space? (C) Option C: Space Exploration & Astrophysics. Discuss the following with your counselor: (a) Colonizing Mars. How can physics help humans live on other planets? (b) Interstellar Travel. Can we reach other stars with warp drives or solar sails? (D) Option D: Advanced Theories & Future Discoveries. Discuss the following with your counselor: (a) Time Travel. Is it possible under extreme conditions? (b) Artificial Gravity. Can we create Earth-like gravity in space stations?

Official Study & Counselor Discussion Points:

(A) Option A: Unsolved Mysteries of Physics. Discuss the following with your counselor:
(B) Option B: Quantum Mechanics & Computing. Discuss the following with your counselor:
(C) Option C: Space Exploration & Astrophysics. Discuss the following with your counselor:
(D) Option D: Advanced Theories & Future Discoveries. Discuss the following with your counselor:







Scouts BSA Test Lab Resource Guide: Physics
Requirement 9 ? Requirement 9

Complete the survey below to complete the test lab requirements

Official Study & Counselor Discussion Points:

